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Area law for four-dimensional SU(2) lattice Yang–Mills theory at $\beta_W < 2/3$, and the limits of single-site mixing criteria

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Abstract

We prove Wilson's area law for four-dimensional SU(2) lattice Yang–Mills theory with Wilson action $\exp(\frac{\beta_W}{2}\sum_p \Re\,\mathrm{Tr}\,U_p)$ for all $\beta_W < 2/3$. This is the full range asserted without proof by Durhuus and Fröhlich in 1980. The largest range previously established with a complete argument is $\beta_W < 1/6$ (Cao–Nissim–Sheffield 2025). This is also what the Durhuus–Fröhlich argument supports once a normalisation in their SU(2) section is corrected. Combining Levin's 1980 Wasserstein estimate for N-vector models with the Durhuus–Fröhlich reduction would give $\beta_W < 2/(3\sqrt{5}) \approx 0.298$. Royer's 1986 geodesic estimate would give $\beta_W < 1/(3\sqrt{\pi^2/3 - 5/2}) \approx 0.375$. We have not found either combination recorded. The proof combines the Durhuus–Fröhlich reduction to three-dimensional O(4) σ-models in random SO(4) environments with a Wasserstein form of Dobrushin's uniqueness condition. The key estimate bounds the Wasserstein Lipschitz constant of the von Mises–Fisher family on S^3 by the optimal value $1/4$ for fields of norm at most 4. This improves Levin's constant $\sqrt{5}/4$ and Royer's geodesic bound $\frac{1}{2}\sqrt{\pi^2/3 - 5/2} \approx 0.444$. Combined with an r-dependent form of Levin's own estimates, certified in interval arithmetic for $|\eta| \geq 4$, it proves his 1980 conjecture for $N = 4$: the Wasserstein Lipschitz constant equals $1/4$ at all fields. A similar certificate proves it for $N = 3$, the classical Heisenberg model; it comes from a harmonic Neumann flow in the unit ball of $\mathbb{R}^4$ and reduces, by symmetry, to two Bessel-function inequalities that we certify in exact rational arithmetic with explicit tail bounds. We then show that $2/3$, the O(4) mean-field point, is the exact limit of single-site Wasserstein–Dobrushin criteria for these models; that the Bakry–Émery criterion holds uniformly exactly for $\beta_W < 1/6$; that single-site total-variation criteria fail from $\beta_W \approx 0.3963$; and that no argument through the slab criterion can pass $\beta_W = 4G_{\mathbb{Z}^3}(0) \approx 1.011$, beyond which the infrared bound gives long-range order in the three-dimensional O(4) model. Numerically the transition occurs at 0.936. All results are at fixed lattice spacing and strong coupling; nothing here concerns the continuum limit.

Contents

1 Introduction **2**
1.1 Background . . . 2
1.2 Explicit ranges for SU(2) . . . 2
1.3 Main results . . . 3
1.4 Ideas of the proof . . . 4

2 Preliminaries **4**
2.1 The slab criterion 4
2.2 Dobrushin's condition in Wasserstein form: a self-contained proof 5

3 The Wasserstein Lipschitz constant of vMF laws **7**
3.1 The Neumann flow bound 7
3.2 Symmetry reduction and closed forms 7
3.3 Two Bessel inequalities 8

4 Proof of Theorem A **10**

5 Limits of single-site criteria: proof of Theorem C **11**
5.1 (i) Bakry–Émery 11
5.2 (ii) Total variation 11
5.3 (iii) Chordal Wasserstein 12
5.4 (iv) The slab ceiling 12

6 Total variation and mean absolute deviation **12**

7 A remark on the Durhuus–Fröhlich normalisation **13**

8 Discussion **14**

A Reproducibility **14**

1 Introduction

1.1 Background

Wilson's area law, $|\langle W_\ell\rangle| \le Ce^{-c\,\mathrm{area}(\ell)}$ for rectangular loops ℓ, is the lattice criterion for confinement of static quarks [18]. In four dimensions it is rigorously known only at strong coupling. Osterwalder and Seiler [15, Thm. 5.1] proved it for sufficiently small β by a cluster expansion, with a threshold that is not explicit. Durhuus and Fröhlich [7] showed that a ν-dimensional lattice gauge theory is an integral of products of $(\nu-1)$-dimensional σ-models in random external gauge fields, and that uniform clustering of certain two-point functions of these σ-models implies the area law for gauge groups with nontrivial centre. For N-vector models, Levin [14] applied Dobrushin's uniqueness theorem and obtained uniqueness up to the mean-field critical temperature multiplied by a numerical factor. Recent probabilistic work [17, 1, 2, 4] has produced explicit strong-coupling regimes; in particular Cao, Nissim and Sheffield [2] verified the Durhuus–Fröhlich condition by the Bakry–Émery method of Shen, Zhu and Zhu [17].

1.2 Explicit ranges for SU(2)

Throughout, β_W denotes the Wilson coupling, $d\mu \propto \prod_p \exp[\frac{\beta_W}{2}\Re\,\mathrm{Tr}\,U_p]\prod_e dU_e$. In the 't Hooft normalisation $\exp[N\beta\sum_p \Re\,\mathrm{Tr}\,U_p]$ of [17, 2], $\beta_W = N^2\beta = 4\beta$; in the normalisation $\exp[\beta\sum_p \Re\chi(U_p)]$ of [7], with χ the fundamental character, $\beta_W = 2\beta$.

Source	Statement (4D SU(2), Wilson β_W)	Status
Osterwalder–Seiler [15, Thm. 5.1]	area law for small β_W	not explicit
Durhuus–Fröhlich [7], pp. 133, 147	asserted: $\beta_W < 1/3$ (claimed with proof), $\beta_W < 2/3$ ("a more refined estimate", stated without proof)	see Remark 7.1
Durhuus–Fröhlich argument, normalisation of Sect. 5 corrected	$\beta_W < 1/6$	Remark 7.1
Cao–Nissim–Sheffield [2]	$\beta_W < 1/6$ ($\beta < 1/(8(d-1))$)	proved
Levin [14] constant $\sqrt{5}/4$ combined with [7]	$\beta_W < 2/(3\sqrt{5}) \approx 0.298$	derivable; not found recorded
Royer [16] geodesic bound combined with [7]	$\beta_W < 1/(3\sqrt{\pi^2/3 - 5/2}) \approx 0.375$	derivable; not found recorded
Theorem A	$\beta_W < 2/3$	proved here

1.3 Main results

Theorem A (Area law)**.** *Let Λ be a periodic box in $\mathbb{Z}^4$ of side L. If $\beta_W < 2/3$, there are $C_1, C_2 > 0$ depending only on β_W such that $|\langle \frac{1}{2} \operatorname{Tr} U_\ell \rangle| \le C_1 e^{-C_2 \operatorname{area}(\ell)}$ for every rectangular loop ℓ with sides at most $L/2$, uniformly in L.*

For $\eta \in \mathbb{R}^4$ let $\mu_\eta \propto e^{\eta \cdot n} d\sigma(n)$ be the von Mises–Fisher law on S^3 (σ the normalised uniform measure), and equip S^3 with the chordal metric $|n - n'|$. W_1 denotes the Kantorovich–Rubinstein distance, which is defined in Section 2.2.

Theorem B (Wasserstein Lipschitz constant of vMF laws)**.** *For $|\eta_0|, |\eta_1| \le 4$, $W_1(\mu_{\eta_0}, \mu_{\eta_1}) \le \frac{1}{4}|\eta_0 - \eta_1|$. The constant $\frac{1}{4}$ is attained as $\eta_0, \eta_1 \to 0$.*

Remark 1.1 (Relation to Levin (1980)). Levin [14, Thm. 3] proved $W_1(\mu_h, \mu_{h'}) \le (\sqrt{5}/N)|h - h'|$ on S^{N-1} for all fields. He bounds the transverse response by $1/N$ (a reflection argument together with an inequality of Dyson–Lieb–Simon [6]) and the longitudinal response by $2/N$, and combines the two by the triangle inequality. He conjectured [14, Rem. 3.7] that $\sqrt{5}$ can be replaced by 1, equivalently that $\sup_{\|f\|_L = 1} |\mu_g[(x \cdot \hat{c} - \langle x \cdot \hat{c} \rangle_g) f]|$, which is our $L(g, \hat{c})$, is maximal at $g = 0$. Theorem B with Corollary 3.6 proves this for $N = 4$ and $|g| \le 4$, and Theorem 3.8 below proves it for $N = 4$ at all fields. The Neumann-flow argument does not split the response into parallel and transverse parts: it controls the quadratic form $Q = c^2 A + s^2 B$ directly, and this is what removes the factor $\sqrt{5}$.

Royer [16] studied the same question with the geodesic metric on S^{N-1}. He proved Dobrushin uniqueness for couplings below $\sqrt{N(N-3)}$, and by a variant better values for $N = 3, 4$. For $N = 4$ his Theorem 3.6 gives the Lipschitz-variance constant $C(3) < \pi^2/3 - 5/2$. With his covariance argument (Lemma 1.2) this yields the geodesic response bound $L_{\mathrm{R}} \le \frac{1}{2}\sqrt{\pi^2/3 - 5/2} = 0.44437\ldots$, that is, a Dobrushin coupling bound of $2.2504\ldots$, matching his table and improving Levin's $4/\sqrt{5} \approx 1.79$. Used in place of Theorem B, it would give $\beta_W < 1/(3\sqrt{\pi^2/3 - 5/2}) \approx 0.37507$ by the slab route and $\beta_W < 1/(9\sqrt{\pi^2/3 - 5/2}) \approx 0.12502$ directly on links. Theorems A and [13] exceed these by the factor $2\sqrt{\pi^2/3 - 5/2} \approx 1.7775$. We have not found either application recorded. He suggested that $\sqrt{N(N-1)}$ (for $N = 4$: ≈ 3.46) might be attainable. With the geodesic metric the mean-field value cannot be reached for $N = 4$. At zero field the test function $\theta = \arccos(v \cdot n)$ gives $\operatorname{Cov}(\theta, \cos\theta) = -8/(9\pi)$, so the geodesic Lipschitz constant is at least $8/(9\pi) > 1/4$, and every single-site geodesic criterion is capped at coupling $9\pi/8 \approx 3.53 < 4$. The chordal metric, which is Levin's original choice, is therefore essential for Theorem 3.8.

Theorem C (Limits of single-site criteria)**.** *For the slab O(4) models of Section 2 with coupling $K = \beta_W$ and arbitrary SO(4) environments:*

(i) the Bakry–Émery criterion, for the product round metric, holds uniformly iff $K < 1/6$;

(ii) every single-site total-variation Dobrushin criterion fails for $K \geq K_{TV} = 0.39629095688\ldots$, the root of $6\mathbf{L}_1(K) = I_1(K)$ ($\mathbf{L}_1$ the modified Struve function);

(iii) single-site chordal-W_1 Dobrushin criteria hold for $K < 2/3$ and fail for $K \geq 2/3$, the O(4) mean-field point $n/z = 4/6$;

(iv) the hypothesis of the slab criterion fails for $K > 4G_{\mathbb{Z}^3}(0) = 1.0109\ldots$, where $G_{\mathbb{Z}^3}$ is the Green's function of the lattice Laplacian. Numerically it fails already for $K > K_c = 0.9360(1)$ [11].

We also prove partial results on the mean absolute deviation of vMF laws (Proposition 6.1) and state a conjecture (Conjecture 6.2) that would extend the unconditional total-variation range $\beta_W < 1/3$ to $\beta_W < \pi/8$. Neither is needed for Theorem A.

1.4 Ideas of the proof

(1) The slab reduction turns the area law into uniform exponential decay of two-point functions of O(4) σ-models with arbitrary SO(4) couplings. (2) A Wasserstein Dobrushin condition reduces this to a single-site Lipschitz constant, Theorem B. (3) A harmonic Neumann flow in B^4 bounds that constant by $\frac{1}{2}\sqrt{Q}$, where Q is a Dirichlet-to-Neumann quadratic form in the density of μ_η. (4) Symmetry splits $Q = c^2A + s^2B$. (5) Degree splitting and closed forms reduce $A, B < 1/4$ to two Bessel inequalities. (6) These are certified in exact rational arithmetic. The Neumann-flow bound is tight at $\eta = 0$, which is why the result reaches the mean-field point.

2 Preliminaries

2.1 The slab criterion

Condition on the horizontal links of two adjacent time slices. The vertical links then form a 3D σ-model on the slab torus with boundary fields A, B,

$$S_{A,B}(Q) = \tfrac{\beta_W}{2} \sum_{e=(x,y)} \Re\,\mathrm{Tr}(Q_x A_e Q_y^{-1} B_e^{-1}).$$

Theorem 2.1 ([7], Thms. 1.2–1.3; as stated in [2], Thm. 2.3)**.** *Let $G \subset U(N)$ be compact and contain zI for some $z \in U(1)$, $z \neq 1$. Suppose that for all boundary fields, all x, y and all matrix indices, $|\operatorname{Cov}_{A,B}((Q_x)_{ij}, (Q_y^{-1})_{kl})| \leq C_1 e^{-C_2 d(x,y)}$ with constants depending only on G, d, β. Then the corresponding lattice Yang–Mills theory satisfies the area law, with constants depending only on G, d, β, for rectangular loops with sides at most $L/2$.*

Lemma 2.2 (SO(4) reduction)**.** *Write $Q = n^0 I + i\,\mathbf{n}\cdot\boldsymbol{\sigma}$ with $n = (n^0, \mathbf{n}) \in S^3$. For $A, B \in SU(2)$ there is $R \in SO(4)$ with $\Re\,\mathrm{Tr}(Q_x A Q_y^{-1} B^{-1}) = 2\,n_x \cdot R n_y$. Hence $S_{A,B} = K \sum_{\langle xy\rangle} n_x \cdot R_{xy} n_y$ with $K = \beta_W$, and the conditional law at x is μ_{η_x} with $\eta_x = K \sum_{y\sim x} R_{xy} n_y$, $|\eta_x| \leq 6K$. The entries of Q_x and Q_x^{-1} are linear in n_x with coefficient vectors of norm at most 1.*

Proof. For $P, Q \in SU(2)$, $\Re\,\mathrm{Tr}(PQ^{-1}) = \Re\,\mathrm{Tr}(PQ^*) = 2\,p \cdot q$. Since $Q_x A Q_y^{-1} B^{-1} = Q_x (B Q_y A^{-1})^{-1}$, we get $\Re\,\mathrm{Tr}(Q_x A Q_y^{-1} B^{-1}) = 2\,n_x \cdot R n_y$, where R is the map sending the quaternion of Q to that of BQA^{-1}; this map is linear and orthogonal, with determinant 1 by connectedness of $SU(2) \times SU(2)$. The last claim is read off from the parametrisation. □

2.2 Dobrushin's condition in Wasserstein form: a self-contained proof

Throughout this subsection S is a countable index set, $(X_i, d_i)_{i\in S}$ are compact metric spaces with $\operatorname{diam} X_i \le D$, and $\Omega = \prod_i X_i$ carries the product topology and its Borel σ-algebra. For $\omega \in \Omega$, $i \in S$ and $x \in X_i$, $\omega^{i,x}$ denotes ω with its i-th coordinate replaced by x. A *single-site system* is a family of probability kernels $\mu_i(dx|\omega)$ on X_i, defined for every ω, measurable in ω and depending only on $\omega_{S\setminus\{i\}}$. Its heat-bath operators act on bounded measurable functions by

$$P_i h(\omega) = \int_{X_i} h(\omega^{i,x})\, \mu_i(dx|\omega).$$

A probability measure μ on Ω is *compatible* if, for each i, P_i is a version of $\mathbb{E}_\mu[\,\cdot\,|\mathcal{F}_{S\setminus\{i\}}]$. Every Gibbs measure of a specification whose one-point kernels are the μ_i is compatible, by the DLR equation for one-point volumes. For bounded h and $j \in S$ set

$$\delta_j(h) = \sup\Big\{ \frac{|h(\omega) - h(\omega')|}{d_j(\omega_j, \omega'_j)} : \ \omega = \omega' \text{ off } j,\ \omega_j \neq \omega'_j \Big\},$$

and define the interaction matrix and its row-sum norm by

$$C_{ij} = \sup\Big\{ \frac{W_1(\mu_i(\cdot|\omega), \mu_i(\cdot|\omega'))}{d_j(\omega_j, \omega'_j)} : \ \omega = \omega' \text{ off } j,\ \omega_j \neq \omega'_j \Big\}\ (i \neq j), \qquad C_{ii} = 0, \qquad \alpha = \sup_i \sum_j C_{ij}.$$

Here W_1 is the Kantorovich–Rubinstein distance,

$$W_1(\rho, \rho') = \sup\Big\{ \int \varphi\, d\rho - \int \varphi\, d\rho' : \ \operatorname{Lip}(\varphi) \le 1 \Big\}.$$

By definition and scaling,

$$\Big| \int \varphi\, d\rho - \int \varphi\, d\rho' \Big| \le \operatorname{Lip}(\varphi)\, W_1(\rho, \rho'). \tag{1}$$

By the Kantorovich–Rubinstein duality theorem, $W_1(\rho, \rho')$ also equals $\inf_\pi \int d\, d\pi$ over couplings π of ρ and ρ'. This paper never uses that fact: every upper and lower bound on W_1 below is proved directly from the definition. No optimal couplings, measurable selections or stochastic processes are needed. When S is infinite we assume that each row of C has finitely many nonzero entries.

Lemma 2.3 (One-site propagation)**.** *Let h be bounded and measurable. Then $\delta_i(P_i h) = 0$; for $j \neq i$, $\delta_j(P_i h) \le \delta_j(h) + C_{ij}\, \delta_i(h)$; and $|h - P_i h| \le D\, \delta_i(h)$ pointwise.*

Proof. $P_i h$ does not depend on ω_i. Let ω, ω' differ only at $j \neq i$. Then

$$P_i h(\omega) - P_i h(\omega') = \int \big[h(\omega^{i,x}) - h(\omega'^{i,x})\big] \mu_i(dx|\omega) + \Big[\int h(\omega'^{i,x}) \mu_i(dx|\omega) - \int h(\omega'^{i,x}) \mu_i(dx|\omega') \Big].$$

The configurations $\omega^{i,x}$ and $\omega'^{i,x}$ differ only at j, so the first term is at most $\delta_j(h)\, d_j(\omega_j, \omega'_j)$. The map $x \mapsto h(\omega'^{i,x})$ has Lipschitz constant at most $\delta_i(h)$, so by (1) the second term is at most $\delta_i(h)\, W_1(\mu_i(\cdot|\omega), \mu_i(\cdot|\omega')) \le \delta_i(h)\, C_{ij}\, d_j(\omega_j, \omega'_j)$. Finally, $|h(\omega) - P_i h(\omega)| \le \int |h(\omega) - h(\omega^{i,x})|\, \mu_i(dx|\omega) \le \delta_i(h)\, D$. □

Lemma 2.4 (Semigroup propagation)**.** *Let $\Lambda \subset S$ be finite and $L_\Lambda = \sum_{i\in\Lambda}(P_i - I)$. This is a bounded operator on bounded measurable functions with the supremum norm, $\|L_\Lambda\| \le 2|\Lambda|$. Let $P_t^\Lambda = e^{tL_\Lambda}$, let Π_Λ be the diagonal matrix with entries $\mathbf{1}_\Lambda$, and let $M_t^\Lambda = \exp(-t\Pi_\Lambda(I-C))$. Then M_t^Λ has nonnegative entries, and for every bounded measurable h with $\sum_j \delta_j(h) < \infty$,*

$$\delta_j(P_t^\Lambda h) \le \sum_k \delta_k(h)\,[M_t^\Lambda]_{kj} \qquad (j \in S,\ t \ge 0).$$

Proof. Regard $\delta(h) = (\delta_j(h))_j$ as a row vector. Let A_i be the identity matrix with its i-th row replaced by the i-th row of C; then $A_i \ge 0$ entrywise, and Lemma 2.3 reads $\delta(P_i h) \le \delta(h)A_i$. Because the A_i are nonnegative, iteration gives $\delta(P_{i_1}\cdots P_{i_m}h) \le \delta(h)A_{i_m}\cdots A_{i_1}$. Summing over all words (using that each δ_j is subadditive and absolutely homogeneous) gives $\delta\big((\sum_{i\in\Lambda} P_i)^m h\big) \le \delta(h)\big(\sum_{i\in\Lambda} A_i\big)^m$, with $\sum_{i\in\Lambda} A_i = |\Lambda|I - \Pi_\Lambda + \Pi_\Lambda C$.

Now $P_t^\Lambda = e^{-|\Lambda|t}\sum_{m\ge0}\frac{t^m}{m!}\big(\sum_{i\in\Lambda} P_i\big)^m$ converges uniformly, and δ_j is lower semicontinuous under pointwise convergence. Hence

$$\delta(P_t^\Lambda h) \le \delta(h)\,e^{-|\Lambda|t}\exp\big(t(|\Lambda|I - \Pi_\Lambda + \Pi_\Lambda C)\big) = \delta(h)\,M_t^\Lambda.$$

Finally, $I - \Pi_\Lambda(I-C)$ has nonnegative entries (recall $C_{ii} = 0$), so $M_t^\Lambda = e^{-t}\exp\big(t(I - \Pi_\Lambda(I-C))\big) \ge 0$. Under the finite-row assumption only finitely many entries of $\Pi_\Lambda C$ are nonzero, so all matrices act nontrivially on a finite set of indices. □

Proposition 2.5 (Wasserstein–Dobrushin covariance estimate)**.** *Let S be finite, μ compatible, and $\alpha < 1$. Then for all bounded measurable f, g,*

$$|\operatorname{Cov}_\mu(f,g)| \le D^2 \sum_{i,k\in S} \delta_k(f)\,[(I-C)^{-1}]_{ki}\,\delta_i(g).$$

Moreover, let ρ be a metric on S and $m \ge 0$ with $\alpha_m := \max_i \sum_j C_{ij}e^{m\rho(i,j)} < 1$. Then $[(I - C)^{-1}]_{ki} \le e^{-m\rho(k,i)}/(1-\alpha_m)$. Also $[(I-C)^{-1}]_{ki} \le \alpha^{\operatorname{dist}_C(k,i)}/(1-\alpha)$, where dist_C is the graph distance for the edges $\{C_{ij} \ne 0\}$.

Proof. Write $L = L_S$ and $P_t = e^{tL}$. We may assume that all $\delta_k(f)$ are finite.

(a) *Invariance and reversibility.* Since $P_i = \mathbb{E}_\mu[\,\cdot\,|\mathcal{F}_{S\setminus\{i\}}]$, we have $\mu(P_i h) = \mu(h)$ and $\mu(g\,P_i h) = \mu(P_i g\,P_i h)$. Hence $\mu(Lh) = 0$ and $\mu(P_t h) = \mu(h)$ for all t.

(b) *Convergence.* Changing coordinates one at a time gives $\operatorname{osc}(h) \le D\sum_j \delta_j(h)$. By Lemma 2.4 with $\Lambda = S$, $\delta(P_t f) \le \delta(f)e^{-t(I-C)}$. The matrix $e^{-t(I-C)} = e^{-t}e^{tC}$ is nonnegative with row sums at most $e^{-(1-\alpha)t}$. So $\operatorname{osc}(P_t f) \le De^{-(1-\alpha)t}\sum_k \delta_k(f)$, and with (a), $P_t f \to \mu(f)$ uniformly.

(c) *Representation.* In operator norm, $\frac{d}{dt}P_t f = LP_t f$. Also $\|Lh\|_\infty \le \sum_i \|P_i h - h\|_\infty \le D\sum_i \delta_i(h)$ by Lemma 2.3. With Lemma 2.4 and (b) this gives $\|LP_t f\|_\infty \le D\sum_i \delta_i(P_t f) \le De^{-(1-\alpha)t}\sum_k \delta_k(f)$, which is integrable on $[0,\infty)$. Hence $f - \mu(f) = -\int_0^\infty LP_t f\,dt$ as a uniformly convergent integral, and

$$\operatorname{Cov}_\mu(f,g) = \mu\big(g\,(f-\mu(f))\big) = -\int_0^\infty \mu(g\,LP_t f)\,dt.$$

(d) *Local form.* For bounded g, h, (a) gives $\mu\big(g(P_i h - h)\big) = \mu(P_i g\,P_i h) - \mu(gh) = -\mu\big((g - P_i g)(h - P_i h)\big)$. So

$$\mu(g\,Lh) = -\sum_i \mu\big((g-P_i g)(h-P_i h)\big), \qquad |\mu\big((g-P_i g)(h-P_i h)\big)| \le D^2\delta_i(g)\,\delta_i(h),$$

the last bound by Lemma 2.3.

(e) *Assembly.* By (c), (d) and Lemma 2.4,

$$|\operatorname{Cov}_\mu(f,g)| \le D^2 \sum_i \delta_i(g) \int_0^\infty \delta_i(P_t f)\, dt \le D^2 \sum_{i,k} \delta_k(f)\, \delta_i(g) \int_0^\infty [e^{-t(I-C)}]_{ki}\, dt.$$

This equals the claimed bound, since $\int_0^\infty e^{-t(I-C)}dt = (I-C)^{-1}$: indeed $(I-C)\int_0^T e^{-t(I-C)}dt = I - e^{-T(I-C)} \to I$.

(f) *Decay.* $(I-C)^{-1} = \sum_{n\ge 0} C^n$. The entry $[C^n]_{ki}$ is a sum over paths $k = j_0 \to j_1 \to \cdots \to j_n = i$ of $\prod_r C_{j_{r-1}j_r}$. By the triangle inequality, $e^{m\rho(k,i)} \le \prod_r e^{m\rho(j_{r-1},j_r)}$, so $[C^n]_{ki}\, e^{m\rho(k,i)} \le [\widetilde{C}^n]_{ki} \le \alpha_m^n$, where $\widetilde{C}_{ij} = C_{ij}e^{m\rho(i,j)}$ has row sums at most α_m. Summing over n gives the weighted bound. For the graph-distance form, $[C^n]_{ki} = 0$ when $n < \operatorname{dist}_C(k,i)$, and $[C^n]_{ki} \le \alpha^n$. □

Remark 2.6 (Attribution and scope). The method goes back to Dobrushin's uniqueness criterion [5]. The covariance estimate in total-variation form, with oscillations in place of the δ_j and constant $1/4$, is due to Föllmer [8]; decay of correlations under Dobrushin's condition goes back to Gross and to Künsch [12]. A recent coupling-based restatement of Dobrushin's theorem in Wasserstein form is [9]. Chatterjee [3] gives a related total-variation argument. The proofs above are the Wasserstein analogues and are self-contained: besides elementary facts about bounded operators and matrices, they use only (1), the tower property of conditional expectations, and (for uniqueness) the Stone–Weierstrass theorem. Numerical tests of these statements on exactly solvable systems are recorded in the appendix as consistency checks (script `SI_YangMills_reproduce_all.py`, Section S5); no proof in this paper depends on them.

3 The Wasserstein Lipschitz constant of vMF laws

For a unit vector v let $L(\eta, v) = \sup\{\operatorname{Cov}_{\mu_\eta}(f, v\cdot n) : f \text{ 1-Lipschitz}\}$. Put $G = (v\cdot n - \mathbb{E}_\eta v \cdot n)e^{\eta\cdot n}/Z(\eta)$, so that $\operatorname{Cov}_{\mu_\eta}(f, v\cdot n) = \int fG\, d\sigma$ and $\int G\, d\sigma = 0$. Let Π_k be the orthogonal projection onto spherical harmonics of degree k in $L^2(\sigma)$ and $Q(\eta, v) = \sum_{k\ge 1} \frac{1}{k}\|\Pi_k G\|^2_{L^2(\sigma)}$.

3.1 The Neumann flow bound

Theorem 3.1. *$L(\eta, v) \le \frac{1}{2}\sqrt{Q(\eta, v)}$, with equality at $\eta = 0$, where both sides equal $1/4$.*

Proof. Let f be 1-Lipschitz on S^3 and F a 1-Lipschitz extension to $\mathbb{R}^4$ (McShane). Let $g = G/|S^3|$ be the density of $G\, d\sigma$ with respect to surface measure dS ($|S^3| = 2\pi^2$), $g = \sum_{k\ge 1} g_k$ its harmonic expansion, and $\Phi = \sum_k r^k g_k/k$, harmonic in B^4 with $\partial_\nu \Phi = g$. By Green's identity (after mollifying F),

$$\int_{S^3} fG\, d\sigma = \int_{S^3} Fg\, dS = \int_{B^4} \nabla F\cdot\nabla\Phi \le \int_{B^4} |\nabla\Phi| \le |B^4|^{1/2}\|\nabla\Phi\|_{L^2(B^4)}.$$

Moreover $\|\nabla\Phi\|_2^2 = \int_{S^3} \Phi g\, dS = \sum_k \|g_k\|^2_{L^2(dS)}/k = Q/(2\pi^2)$. Since $|B^4| = \pi^2/2$, the bound is $\sqrt{Q}/2$. At $\eta = 0$, $G = v\cdot n$ is of pure degree 1 with $\|G\|^2 = 1/4$, so $Q = 1/4$; and $f = v\cdot n$ is 1-Lipschitz with $\operatorname{Cov}_\sigma(v\cdot n, v\cdot n) = 1/4$, so $L(0, v) = 1/4$. □

3.2 Symmetry reduction and closed forms

Rotate so that $\eta = re_1$ and $v = ce_1 + se_2$ ($c = \cos\theta$, $s = \sin\theta$). Write $t = n_1$, $F(t) = e^{rt}/Z$, $Z = 2I_1(r)/r$, $a = I_2(r)/I_1(r)$. Then $G = c(t-a)F + s\, n_2 F$.

Proposition 3.2. *$Q = c^2A(r)+s^2B(r)$ with $A = \sum_k \frac{1}{k}\|\Pi_k((t-a)F)\|^2$ and $B = \sum_k \frac{1}{k}\|\Pi_k(n_2F)\|^2$.*

Proof. Let $H \cong SO(3)$ be the stabiliser of e_1. Each space of degree-k harmonics decomposes under H as $\bigoplus_{\ell=0}^{k} V_\ell$ with each V_ℓ occurring once (branching $SO(4) \downarrow SO(3)$). $(t-a)F$ is H-invariant, so its projections lie in the V_0 components. n_2F is odd under $n_2 \mapsto -n_2$, which fixes V_0 pointwise, so its projections are orthogonal to V_0. The cross terms vanish. □

Proposition 3.3. *Let $f(r) = 2I_1(r)/r = \mathbb{E}_\sigma[e^{rt}]$, $\beta_k = \|\Pi_k(n_2F)\|^2$ and $\alpha_k = \|\Pi_k((t-a)F)\|^2$. Using $1/k \le 1/3$ for $k \ge 3$ and Parseval,*

$$B \le \tfrac{2}{3}\beta_1 + \tfrac{1}{6}\beta_2 + \tfrac{1}{3}\|n_2F\|^2, \qquad A \le \tfrac{2}{3}\alpha_1 + \tfrac{1}{6}\alpha_2 + \tfrac{1}{3}\|(t-a)F\|^2,$$

where

$$\beta_1 = 4\Big(\frac{I_2}{rI_1}\Big)^2, \quad \beta_2 = 24\Big(\frac{I_3}{rI_1}\Big)^2, \quad \|n_2F\|^2 = \frac{I_2(2r)}{8I_1(r)^2},$$
$$\alpha_1 = 4\frac{(f''f - f'^2)^2}{f^4}, \quad \alpha_2 = 16\frac{(f'''f - f''f')^2}{f^4}, \quad \|(t-a)F\|^2 = \frac{f''(2r)f^2 - 2ff'f'(2r) + f'^2f(2r)}{f^4}.$$

Proof. The first inequality is $B = \beta_1 + \frac{1}{2}\beta_2 + \sum_{k\ge3}\beta_k/k \le \beta_1 + \frac{1}{2}\beta_2 + \frac{1}{3}(\|n_2F\|^2 - \beta_1 - \beta_2)$, using $\Pi_0(n_2F) = 0$; similarly for A, using $\Pi_0((t-a)F) = \mathbb{E}_\eta[t-a] = 0$. The degree-1 and degree-2 pieces are spanned by n_2 and n_1n_2 (odd part) and by $U_1(t) = 2t$, $U_2(t) = 4t^2-1$ (zonal), with $\|n_2\|^2 = 1/4$, $\|n_1n_2\|^2 = 1/24$, $\|U_k(t)\| = 1$. The closed forms follow from $\mathbb{E}[n_2^2|t] = (1-t^2)/3$, the law $\frac{2}{\pi}\sqrt{1-t^2}\,dt$ of t, $\mathbb{E}_\sigma[t^je^{rt}] = f^{(j)}(r)$, and the recurrences $(I_1/r)' = I_2/r$ and $(2I_2/r^2)' = 2I_3/r^2$. □

3.3 Two Bessel inequalities

Multiplying $\frac{1}{4} - B_{\text{bound}}$ by $24r^2I_1^2$ and $\frac{1}{4} - A_{\text{bound}}$ by $12f^4$ gives the functions

$$F_B(r) = 6r^2I_1(r)^2 - 64I_2(r)^2 - 96I_3(r)^2 - r^2I_2(2r),$$
$$D_A(r) = 3f^4 - 32(f''f - f'^2)^2 - 32(f'''f - f''f')^2 - 4\big[f''(2r)f^2 - 2ff'f'(2r) + f'^2f(2r)\big].$$

Lemma 3.4. *For $0 < r \le 4$: $F_B(r)/r^8 > 8.466 \times 10^{-4}$ and $D_A(r)/r^2 > 0.3749$. Consequently the bounds of Proposition 3.3 for A and B are strictly less than $1/4$ on $(0, 4]$.*

Certificate. Both functions are power series in r^2 with exactly computable rational coefficients.

F_B. Write $F_B = \sum_k c_kr^{2k}$. Then $c_0 = \cdots = c_3 = 0$, $c_4 = \frac{1}{1152}$, $c_5 = \frac{7}{92160}$, $c_6 = \frac{1}{368640}$, $c_7 = \frac{17}{371589120}$ and $c_8 = \frac{1}{9909043200}$, all positive; $c_9 < 0$. For $r \le 4$,

$$F_B \ge r^8\Big(c_4 - \sum_{k\ge9}|c_k|16^{k-4}\Big).$$

The exact rational sum over $9 \le k \le 80$ is $2.1438\ldots \times 10^{-5}$, so $c_4 - \sum_{9\le k\le80} = 8.46617\ldots \times 10^{-4}$. For $k \ge 81$, $|c_k|$ is bounded by the coefficient of the nonnegative-coefficient majorant $M_B(r) = 6r^2I_1^2 + 64I_2^2 + 96I_3^2 + r^2I_2(2r)$, and $C_kR^{2k} \le M_B(R)$ with $R = 8$. We bound $M_B(8)$ by a rational number using $I_\nu(x) \le (x/2)^\nu e^x/\nu!$ and $e < 2.7183$. This gives a tail at most 5.4×10^{-43}.

D_A. Write $D_A = \sum_k d_kr^{2k}$. Then $d_0 = 0$, $d_1 = \frac{3}{8}$, $d_2 = \frac{9}{64}$, $d_3 = \frac{143}{5760}$, and the exact coefficients satisfy $d_k \ge 0$ for all $1 \le k \le 189$. The tail $k \ge 190$ is bounded by the majorant obtained by replacing every minus sign in D_A by a plus sign (all of f, f', f'', f''' have nonnegative coefficients), This majorant is bounded rationally in the same way, via $f' = 2I_2/r$, $f'' = 2I_2/r^2 + 2I_3/r$ and $f''' = 6I_3/r^2 + 2I_4/r$. It gives at most 10^{-98} for D_A/r^2 when $r \le 4$. Hence $D_A/r^2 \ge \frac{3}{8} - 10^{-98}$. The whole certificate uses exact rational arithmetic only.

Finally $\frac{1}{4} - B_{\text{bound}} = F_B/(24r^2I_1^2) > 0$ and $\frac{1}{4} - A_{\text{bound}} = D_A/(12f^4) > 0$. □

Remark 3.5. The coefficient computations are reproducible with the script `cert_series.py` and the rational tail bounds with `cert_tails.py`. The bounds stated in the lemma have also been obtained independently by an external reader using a separate implementation. A direct 50-digit evaluation on the grid $r = k/200$ gives $\min F_B/r^8 = 8.6806 \times 10^{-4}$ and $\min D_A/r^2 = 0.3750035$, both at $r = 0.005$, consistent with the certificates.

Corollary 3.6. *For $|\eta| \le 4$ and unit v: $L(\eta, v) \le \frac{1}{2}\sqrt{\max(A, B)} < \frac{1}{4}$ for $\eta \neq 0$, and $L(0, v) = \frac{1}{4}$.*

Proof of Theorem B. Let $\eta_t = \eta_0 + t\delta$, $\delta = \eta_1 - \eta_0$; the segment lies in the ball of radius 4. By the definition of W_1 (Section 2.2) and $\frac{d}{dt}\int f\, d\mu_{\eta_t} = \mathrm{Cov}_{\mu_{\eta_t}}(f, \delta \cdot n)$,

$$W_1(\mu_{\eta_0}, \mu_{\eta_1}) = \sup_f \int_0^1 \mathrm{Cov}_{\mu_{\eta_t}}(f, \delta \cdot n)\, dt \le |\delta| \sup_t L(\eta_t, \hat{\delta}) \le |\delta|/4.$$

Attainment: $W_1(\mu_0, \mu_\delta) \ge \int (\hat{\delta} \cdot n)\, d(\mu_\delta - \mu_0) = |\delta|\, \mathrm{Var}_\sigma(\hat{\delta} \cdot n) + o(|\delta|) = |\delta|/4 + o(|\delta|)$. □

Corollary 3.7 (Levin's mean-field conjecture for $N = 4$ in zero external field)**.** *Consider O(4) spins $\sigma_a \in S^3$ on $\mathbb{Z}^\nu$ with arbitrary two-body interactions $\sum_{\{a,b\}} \sigma_a \cdot J_{ab}\sigma_b$, summed over unordered pairs, where the J_{ab} are 4×4 real matrices with $J_{ba} := J_{ab}^{\mathsf{T}}$, and arbitrary one-body terms $\sum_a h_a \cdot \sigma_a$. This is the convention of [14]. If $\sup_a \sum_b \|J_{ab}\|_{op} < 4$, the Gibbs state is unique. The constant 4 is the sharp threshold of single-site chordal Dobrushin criteria (at zero field the response equals $1/4$). In the isotropic ferromagnetic case it coincides with the mean-field critical temperature. It is not a statement about the location of any phase transition. In zero external field the proof needs only Theorem B; with external fields it uses Theorem 3.8.*

Proof. With this convention the conditional law at a is μ_{g_a} with $g_a = h_a + \sum_{b\neq a} J_{ab}\sigma_b$. With an ordered double sum the interaction part would instead be $\sum_b (J_{ab} + J_{ba}^{\mathsf{T}})\sigma_b$. Changing σ_b moves g_a along a segment by at most $\|J_{ab}\|_{op}|\sigma_b - \sigma_b'|$. For $h \equiv 0$ the segment lies in the ball of radius 4, and Theorem B applies. For general h, Theorem 3.8 applies. In both cases the Dobrushin coefficient is at most $\frac{1}{4}\|J_{ab}\|_{op}$, so the row sums are less than 1. Dobrushin's uniqueness theorem [5] (see also [14, Thm. 2]) applies. □

Theorem 3.8 (Levin's conjecture for $N = 4$)**.** *For all $g \in \mathbb{R}^4 \setminus \{0\}$ and unit $\hat{c}$, $L(g, \hat{c}) < \frac{1}{4} = L(0, \hat{c})$. Consequently $W_1(\mu_h, \mu_{h'}) \le \frac{1}{4}|h - h'|$ for all $h, h' \in \mathbb{R}^4$.*

Proof. For $0 < |g| \le 4$ this is Corollary 3.6. Let $r = |g| \ge 4$, $a = a(r) = I_2(r)/I_1(r) = \mathbb{E}_g[t]$ with $t = \hat{g} \cdot n$, and $\mathrm{Var}(t) = a'(r) = 1 - a^2 - 3a/r$ (Riccati equation for I_2/I_1).

Transverse part (Levin [14, Lemma 3.2]). For a unit $e \perp g$ and f 1-Lipschitz, the reflection R with $x \cdot e \mapsto -x \cdot e$ preserves μ_g and satisfies $|f(x) - f(Rx)| \le 2|x \cdot e|$. Hence $|\mu_g[(x \cdot e)f]| = \frac{1}{2}|\mu_g[(x \cdot e)(f - f \circ R)]| \le \mu_g[(x \cdot e)^2] = a/r$.

Longitudinal part (Levin [14, Lemma 3.3, (3.19)–(3.24)]). Let $\hat{f}(t)$ be the average of f over the 2-sphere $\{\hat{g} \cdot n = t\}$. Writing $t = \cos\theta$, points with the same angular coordinates satisfy $|x - x'|/|t - t'| = 1/\sin\frac{\theta+\theta'}{2} \le \sqrt{2/(1 - t')}$ when $t' \ge 0$. Hence $|\hat{f}(t) - \hat{f}(a)| \le \sqrt{2/(1-a)}\,|t - a|$ (note $a > 0$), and $|\mu_g[(t - a)f]| = |\mu_g[(t - a)(\hat{f}(t) - \hat{f}(a))]| \le \sqrt{2/(1-a)}\ \mathrm{Var}(t)$.

Combination. Write $\hat{c} = c_\parallel \hat{g} + c_\perp e$ with $c_\parallel^2 + c_\perp^2 = 1$. Then

$$L(g, \hat{c}) \le |c_\parallel|\sqrt{2/(1-a)}\ \mathrm{Var}(t) + |c_\perp|\, a/r \le \Lambda(r) := \left(\tfrac{2\,\mathrm{Var}(t)^2}{1-a} + \tfrac{a^2}{r^2}\right)^{1/2}.$$

Certificate that $\Lambda(r) < \frac{1}{4}$ for $r \ge 4$. Since $a' = \mathrm{Var}(t) > 0$, a is increasing. On a cell $[r_j, r_{j+1}]$ we therefore have $a \in [a(r_j), a(r_{j+1})]$ and $\mathrm{Var}(t) \le 1 - a(r_j)^2 - 3a(r_j)/r_{j+1}$. This gives an upper bound

for Λ^2 on the cell in terms of $a(r_j)$ and $a(r_{j+1})$ only. We evaluated these enclosures by the power series of I_1 and I_2 in outward-rounded interval arithmetic, with geometric tail bounds, at 59 dyadic points of $[4, 262.45]$. The largest resulting cell bound on Λ^2 is $0.06237 < 1/16$. For $r \geq R = 262.45$ we use $\mathrm{Var}(t) \leq \mathbb{E}(1-t)^2 \leq 2(1-a)$, giving $\Lambda^2 \leq 8(1-a(R)) + R^{-2} \leq 0.0457 < 1/16$. The Wasserstein statement follows by integrating along segments, as in the proof of Theorem B. □

Theorem 3.9 (Levin's conjecture for $N = 3$)**.** *On S^2, for all $g \in \mathbb{R}^3 \setminus \{0\}$ and unit $\hat{c}$, $L(g, \hat{c}) < \frac{1}{3} = L(0, \hat{c})$. Consequently $W_1(\mu_h, \mu_{h'}) \leq \frac{1}{3}|h - h'|$ for all $h, h' \in \mathbb{R}^3$. The classical Heisenberg model with arbitrary pair interactions (in the convention of Corollary 3.7) and arbitrary one-body terms therefore has a unique Gibbs state whenever $\sup_a \sum_b \|J_{ab}\|_{op} < 3$, which is the sharp single-site chordal Dobrushin threshold.*

Proof. Weak fields, $0 < r \leq \frac{7}{2}$. The proof of Theorem 3.1 applies verbatim on S^2 with the constant $(|B^3|/|S^2|)^{1/2} = 3^{-1/2}$, giving $L \leq \sqrt{Q/3}$. As in Proposition 3.2, $Q = c^2 A + s^2 B$ (the stabiliser of $\hat{g}$ is now $\mathrm{SO}(2)$, and in each degree its invariant harmonics are again the single zonal line), where $A = \sum_k \alpha_k/k$ uses the Legendre polynomials $P_k(t)$ and $B = \sum_k \beta_k/k$ uses the harmonics $x_2 P_k'(t)$. Keep the degrees $k = 1, 2$ exactly and use $1/k \leq 1/3$ for $k \geq 3$. This gives $A \leq A_3$ and $B \leq B_3$, with A_3, B_3 rational combinations of $m_j(r) = \frac{1}{2}\int_{-1}^{1} t^j e^{rt} dt = \sum_{n \equiv j \ (2)} r^n/(n!(n+j+1))$ and of $m_j(2r)$.

The functions $(\frac{1}{3} - B_3)m_0^2$ and $(\frac{1}{3} - A_3)m_0^4$ are power series in r with rational coefficients. The coefficients vanish below order 2, equal $1/90$ and $4/45$ at order 2, and are nonnegative through order 120 (exact arithmetic). A majorant with nonnegative coefficients, using $m_j(x) \leq e^x$, $m_1(x)/x \leq e^x$ and $e < 2.7183$ at radius 8, bounds the remainder beyond order 120 by $3 \times 10^{-37} r^2$ and $8 \times 10^{-30} r^2$ for $r \leq \frac{7}{2}$. Hence $A_3, B_3 < \frac{1}{3}$ on $(0, \frac{7}{2}]$.

Strong fields, $r \geq 3.3$. The argument of Theorem 3.8 with $N = 3$ gives $L \leq \Lambda(r) = (2\,\mathrm{Var}(t)^2/(1-a) + a^2/r^2)^{1/2}$. Here $a = \coth r - 1/r$ and $\mathrm{Var}(t) = a' = 1 - a^2 - 2a/r > 0$. Interval arithmetic on 38 cells of $[3.3, 103.53]$, using the monotonicity of a, gives $\Lambda^2 \leq 0.110472 < 1/9$. For $r \geq 103.53$, $\Lambda^2 \leq 8(1-a) + r^{-2} \leq 0.07737$.

The two regimes overlap on $[3.3, 3.5]$. The rational-series and quadrature implementations of A_3 and B_3 agree to 10^{-17}. □

Remark 3.10 (Other N). On S^{N-1} the Neumann-flow bound is $\sqrt{Q/N}$, since $|B^N|^{1/2}/|S^{N-1}|^{1/2} = N^{-1/2}$. It is exact at zero field for every N, because $Q(0, v) = 1/N$. However, for $N \geq 5$ it exceeds $1/N$ at small positive fields. Indeed $B \geq \beta_1 + \beta_2/2$, and this already exceeds $1/N$: for example $0.2003 > 1/5$ at $N = 5$, $r = 0.3$ (computed with 60 and 90 Gegenbauer degrees, which agree). So the method as it stands cannot settle Levin's conjecture for $N \geq 5$. For $N = 3$, the classical Heisenberg model, the same two-regime strategy succeeds (Theorem 3.9).

4 Proof of Theorem A

Let $K = \beta_W < 2/3$. In the slab model of Lemma 2.2, changing n_y from u to u' changes η_x by $\delta = K R_{xy}(u - u')$, with $|\delta| = K|u - u'|$, and every field in question has norm at most $6K < 4$. By Theorem B, $C_{xy} \leq K/4$ for neighbours and $C_{xy} = 0$ otherwise, so $\alpha \leq 6K/4 < 1$. The real and imaginary parts of $(Q_x)_{ij}$ and $(Q_y^{-1})_{kl}$ are 1-Lipschitz functions of n_x, respectively n_y (Lemma 2.2). Applying Proposition 2.5 with $D = 2$ to each of the four real covariances gives

$$|\,\mathrm{Cov}_{A,B}((Q_x)_{ij}, (Q_y^{-1})_{kl})| \leq \frac{16}{1-\alpha}\,\alpha^{d(x,y)},$$

uniformly in the environment and in L. Theorem 2.1 applies because the centre $\{\pm I\}$ of SU(2) is nontrivial. □

5 Limits of single-site criteria: proof of Theorem C

A single-site criterion uses a matrix C_{xy} defined as a supremum, over environments and over configurations differing only at y, of a distance between the conditional laws at x. It succeeds if the spectral radius satisfies $\rho(C) < 1$; by Perron–Frobenius this is equivalent to the existence of positive weights w with $\sum_y C_{xy} w_y \le (1-\varepsilon) w_x$. Let A denote the adjacency matrix of the slab torus, which is 6-regular, so $\rho(A) = 6$.

5.1 (i) Bakry–Émery

On $(S^3)^\Lambda$ with the round metric, $\mathrm{Ric} = 2g$. With $V = -K \sum n_x \cdot R_{xy} n_y$ and using $\mathrm{Hess}(w \cdot n) = -(w \cdot n) g$ on S^3,

$$\mathrm{Hess}\, V(u,u) = K \sum_{\langle xy \rangle} \big[(n_x \cdot R_{xy} n_y)(|u_x|^2 + |u_y|^2) - 2 u_x \cdot R_{xy} u_y \big] \ge -12K |u|^2,$$

by $|n_x \cdot R n_y| \le 1$, $|u_x \cdot R u_y| \le \tfrac{1}{2}(|u_x|^2 + |u_y|^2)$ and six bonds per site. So $\mathrm{Ric} + \mathrm{Hess}\, V \ge (2 - 12K) g$, pointwise in the configuration and uniformly in the environment. Equality is attained with $R \equiv I$ on an even torus, $n_x = (-1)^{x_1+x_2+x_3} e$ and $u_x = w \perp e$: then $n_x \cdot n_y = -1$ and $u_x \cdot u_y = |w|^2$ on every bond. The threshold is unchanged by constant rescalings of the metric. A numerical check on a 4^3 torus gives minimum eigenvalues 0.8, 0 and -1 at $K = 0.1$, $1/6$ and $1/4$ for this configuration, and larger values for random configurations and environments. The condition of [2], $\frac{N+2}{2} - 1 - 4N\beta(d-1) > 0$, is $\beta_W < 1/6$ for SU(2) and $d = 4$; it is therefore exactly sharp for this model.

5.2 (ii) Total variation

Fix a bond xy. Choose the other five neighbours so that their vectors Rn sum to zero (a regular pentagon), so that $\eta = K R_{xy} n_y$ with $|\eta| = K$, and flip $n_y \mapsto -n_y$. The component $t = \hat{\eta} \cdot n$ has density $\frac{2}{\pi}\sqrt{1-t^2}\, e^{Kt}/f(K)$. Using $\int_0^1 \sqrt{1-t^2} \sinh(Kt)\, dt = \pi \mathbf{L}_1(K)/(2K)$,

$$\|\mu_\eta - \mu_{-\eta}\|_{TV} = \frac{\mathbf{L}_1(K)}{I_1(K)}.$$

Hence $C \ge (\mathbf{L}_1/I_1) A$ entrywise and $\rho(C) \ge 6\mathbf{L}_1(K)/I_1(K)$. The ratio $R = \mathbf{L}_1/I_1$ is strictly increasing on $(0,\infty)$, with $R(0^+) = 0$ and $R(\infty) = 1$. So $6R = 1$ has a unique root, and the criterion fails for $K \ge K_{TV} = 0.39629095688184401321\ldots$.

Proof of monotonicity. By Poisson's integrals, with $w(t) = \sqrt{1-t^2}$, we have $I_1(K) = \frac{2K}{\pi}\int_0^1 w \cosh(Kt)\, dt$ and $\mathbf{L}_1(K) = \frac{2K}{\pi}\int_0^1 w \sinh(Kt)\, dt$. Hence $R(K) = \mathbb{E}_\nu[\tanh(Kt)]$, where $\nu = \nu_K \propto w \cosh(Kt)\, dt$ on $[0,1]$. Differentiating,

$$R'(K) = \mathbb{E}_\nu[t \operatorname{sech}^2(Kt)] + \mathrm{Cov}_\nu\big(\tanh(Kt),\, t \tanh(Kt)\big).$$

The first term is strictly positive. The second is nonnegative by Chebyshev's association inequality, because both functions are nondecreasing in $t \in [0,1]$. Finally, $0 \le R \le \tanh K \to 0$ as $K \to 0$, and $R \to 1$ as $K \to \infty$ because ν_K concentrates at $t = 1$. Quadrature reproduces $\mathbf{L}_1/I_1$ to 20 digits, and the ratio equals $1/6$ at K_{TV}.

5.3 (iii) Chordal Wasserstein

The upper bound $C_{xy} \le K/4$ for $K \le 2/3$ was shown in Section 4. For the lower bound, choose the other five neighbours to cancel $R_{xy}n_y$, so that $\eta_x = 0$, and perturb n_y. By the attainment part of Theorem B, $C_{xy} \ge K/4$. Thus $C \ge (K/4)A$ and $\rho(C) \ge 3K/2$, so every such criterion fails for $K \ge 2/3$. Linearising the vMF mean, $a(r) = I_2/I_1 \approx r/4 = r/n$, so the self-consistency equation $m = (Kz/n)m$ identifies $K = n/z = 2/3$ as the mean-field point.

5.4 (iv) The slab ceiling

For the trivial environment $R \equiv I$ the slab model is the standard O(4) model on the torus $(\mathbb{Z}/L)^3$ at $\beta = K$. This environment arises from the trivial horizontal fields. Reflection positivity gives Gaussian domination and hence the infrared bound $\hat{S}(p) \le 4/(K\,E(p))$ for $p \ne 0$, where $\hat{S}(p) = |\Lambda|^{-1}\langle|\sum_x e^{ipx}n_x|^2\rangle$ and $E(p) = 2\sum_\mu(1-\cos p_\mu)$ [10]. Combined with the sum rule $|\Lambda|^{-1}\sum_p \hat{S}(p) = 1$, this gives

$$\frac{1}{|\Lambda|^2}\sum_{x,y}\langle n_x \cdot n_y\rangle = \frac{\hat{S}(0)}{|\Lambda|} \ge 1 - \frac{4G_L(0)}{K}, \qquad G_L(0) = \frac{1}{|\Lambda|}\sum_{p\ne 0}\frac{1}{E(p)} \to G_{\mathbb{Z}^3}(0) = 0.2527310098\ldots$$

Here $G_{\mathbb{Z}^3}(0) = W/6$, with Watson's $W = \frac{\sqrt{6}}{32\pi^3}\Gamma(\frac{1}{24})\Gamma(\frac{5}{24})\Gamma(\frac{7}{24})\Gamma(\frac{11}{24})$. For $K > 4G_{\mathbb{Z}^3}(0)$ the averaged correlation therefore stays bounded below uniformly in L. That is incompatible with uniform exponential decay of the entry covariances, because $n_x \cdot n_y = \frac{1}{2}\Re\,\mathrm{Tr}(Q_xQ_y^{-1})$ is a linear combination of products of matrix entries and $\langle Q_x\rangle = 0$ by symmetry. The statement needs no monotonicity of long-range order in K. The numerical transition at $K_c = 0.9360(1)$ [11] indicates that the slab route in fact stops earlier. This concerns the method, not the area law itself.

6 Total variation and mean absolute deviation

Along a segment, $\|\mu_\eta - \mu_{\eta'}\|_{TV} \le \frac{1}{2}|\delta|\sup_\zeta \mathbb{E}_\zeta|\hat{\delta}\cdot(n - \mathbb{E}_\zeta n)|$, so total-variation criteria are controlled by the mean absolute deviation (MAD) of linear functionals.

Proposition 6.1. *(a) For $|\zeta| \le 4$ and unit v, $\mathbb{E}_\zeta|v\cdot(n-\mathbb{E}_\zeta n)| \le \frac{1}{2}$. Consequently single-site TV criteria succeed for $\beta_W < 1/3$. (b) $\mathbb{E}_0|v\cdot n| = \frac{4}{3\pi}$ for every v. (c) If $v \perp \zeta$, then $\mathbb{E}_\zeta|v\cdot n|$ is nonincreasing in $|\zeta|$, and hence at most $\frac{4}{3\pi}$. (d) (Local maximality.) Write $\zeta = r(\cos\theta\, v + \sin\theta\, w)$ with $w \perp v$ a unit vector. Then, uniformly in v and θ,*

$$\mathbb{E}_\zeta|v\cdot(n-\mathbb{E}_\zeta n)| = \frac{4}{3\pi} - \frac{r^2}{\pi}\Big(\frac{13}{120}\cos^2\theta + \frac{1}{30}\sin^2\theta\Big) + o(r^2).$$

In particular $\zeta = 0$ is a strict local maximum of $\sup_v \mathbb{E}_\zeta|v\cdot(n-\mathbb{E}_\zeta n)|$, so Conjecture 6.2 holds on a neighbourhood of $\zeta = 0$.

Proof. (a) Since $v\cdot n$ is 1-Lipschitz, $\mathrm{Var}_\zeta(v\cdot n) \le L(\zeta, v) \le \frac{1}{4}$ by Corollary 3.6, so the MAD is at most $\frac{1}{2}$. Then $C_{xy} \le \frac{1}{2}\cdot 2K\cdot\frac{1}{2} = K/2$, and the row sum $3K$ is less than 1 for $K < 1/3$.

(b) The law of t under σ is $\frac{2}{\pi}\sqrt{1-t^2}\,dt$, so $\mathbb{E}|t| = \frac{4}{\pi}\int_0^1 t\sqrt{1-t^2}\,dt = \frac{4}{3\pi}$.

(c) Take $\zeta = re_1$, $v = e_2$. The projection (n_1, n_2) has density proportional to e^{rx} on the unit disk, and given x, y is uniform on the chord. So $\mathbb{E}|n_2| = \frac{1}{2}\mathbb{E}_r[h(x)]$ with $h(x) = \sqrt{1-x^2}$, and $\frac{d}{dr}\mathbb{E}_r[h] = \mathrm{Cov}_r(h(x), x)$. Conditioning on $u = |x|$, $\mathbb{E}[x\,|\,u] = u\tanh(ru)$, so $\mathrm{Cov}_r(h, x) = \mathrm{Cov}_u(h(u), u\tanh(ru)) \le 0$ by Chebyshev's association inequality, since h is decreasing and $u\tanh(ru)$ is nondecreasing.

(d) Put $X = v \cdot n$, $F(\zeta, c) = \mathbb{E}_\zeta|X - c|$ and $M(\zeta) = F(\zeta, m(\zeta))$ with $m(\zeta) = \mathbb{E}_\zeta X$.

Smoothness and symmetry. F is real-analytic in ζ for fixed c. In c it satisfies $\partial_c F = 2\,\mathbb{P}_\zeta(X < c) - 1$ and $\partial_c^2 F = 2q_\zeta(c)$, where q_ζ is the density of X under μ_ζ; this density is jointly continuous in (ζ, c). Hence M is C^2. It is even, because $n \mapsto -n$ maps μ_ζ to $\mu_{-\zeta}$ and $X - m$ to $-(X - m)$. Its Hessian at 0 is invariant under rotations fixing v. So $M(\zeta) = \frac{4}{3\pi} - \kappa_\parallel r^2 \cos^2\theta - \kappa_\perp r^2 \sin^2\theta + o(r^2)$, uniformly in v by rotation covariance.

Transverse curvature. For $\theta = \pi/2$, (c) gives $M = \frac{1}{2}\mathbb{E}_r h(x)$ with $x = \hat{\zeta} \cdot n$. Since $\mathbb{E}_r h = \mathbb{E}_0 h + \frac{r^2}{2}\operatorname{Cov}_0(h, x^2) + O(r^4)$, $\mathbb{E}_0 h = \frac{8}{3\pi}$, $\mathbb{E}_0 x^2 = \frac{1}{4}$ and $\mathbb{E}_0[hx^2] = \frac{8}{15\pi}$, we get $\kappa_\perp = -\frac{1}{4}\operatorname{Cov}_0(h, x^2) = \frac{1}{30\pi}$.

Longitudinal curvature. For $\theta = 0$, $M = \mathbb{E}_0[|x - a|e^{rx}]/\mathbb{E}_0[e^{rx}]$ with $a = a(r) = \frac{r}{4} + O(r^3)$. Expand each factor to second order in r:

- $\mathbb{E}_0|x - a| = \frac{4}{3\pi} + a^2 q_0(0) + O(a^4)$, with $q_0(0) = \frac{2}{\pi}$;
- $\mathbb{E}_0[|x - a|x] = -a\,\mathbb{E}_0|x| + O(a^2)$;
- $\mathbb{E}_0[|x|x^2] = \frac{8}{15\pi}$;
- $\mathbb{E}_0 e^{rx} = 1 + \frac{r^2}{8} + O(r^4)$.

Collecting terms, $M = \frac{4}{3\pi} + \frac{r^2}{\pi}\left(\frac{1}{8} - \frac{1}{3} + \frac{4}{15} - \frac{1}{6}\right) + O(r^3)$, so $\kappa_\parallel = \frac{13}{120\pi}$.

Both curvatures are positive, with $\kappa_\perp < \kappa_\parallel$, which gives the stated local maximality. □

Conjecture 6.2. $\sup_{\zeta, v} \mathbb{E}_\zeta|v \cdot (n - \mathbb{E}_\zeta n)| = \frac{4}{3\pi}$. *If true, single-site TV criteria succeed for* $\beta_W < \pi/8 \approx 0.3927$.

By Proposition 6.1(d), the conjecture holds near $\zeta = 0$, with the least curvature in the transverse direction. In 20-digit quadrature on a grid of $r \in [0.05, 10]$ and $\theta \in [0, \pi/2]$, the maximum over θ is always attained at $\theta = \pi/2$, the case proved in Proposition 6.1(c), and never exceeds $4/(3\pi)$. This is evidence, not a proof. A computer-assisted proof would combine (d), made quantitative with an explicit remainder, with interval arithmetic on a compact range of r and the bound MAD $\le \sqrt{\text{Var}}$ for large r. Unconditionally, the TV threshold lies in [1/3, 0.39629]; under Conjecture 6.2 it lies in $[\pi/8, 0.39629]$.

7 A remark on the Durhuus–Fröhlich normalisation

Remark 7.1. Durhuus and Fröhlich define the action $A_\nu^{YM} = -\sum_p \Re\chi(g_{\partial p})$ with χ an irreducible character [7, (1.13)], and the measure $Z^{-1}e^{-\beta A_\nu^{YM}}$ [7, (1.14)]. In their Sect. 5 they take χ to be "the character of the fundamental representation of SU(2)" [7, p. 131].

For the map φ of [7, (5.1)], a direct computation gives $\operatorname{Tr}(\varphi(S_1)^{-1}\varphi(S_2)) = 2\,S_1 \cdot S_2$. Hence the vertical-plaquette part of their action equals $(S, \Delta_{b,t}S) + 2(\nu - 1)|\Lambda^0|$ on unit spins. This agrees with their displayed identity $\sum_{ij \subset \Lambda^0} \operatorname{tr}(g_i^{-1} b_{ij} g_j t_{ij}^{-1}) = (S, \Delta_{b,t}S) + 2(\nu - 1)|\Lambda^0|$ [7, p. 131], whose left side is that part of the action. The induced σ-model weight is therefore proportional to $e^{\beta(S, \Delta_{b,t}S)}$: per-bond coupling $2\beta = \beta_W$.

The σ-model analysed in Sect. 5 is instead $\int e^{\frac{1}{2}\beta(S_t, \Delta S_t)} \prod_i \delta(|(S_t)_i|^2 - 1)\, d(S_t)_i$ [7, p. 133], equivalently their (5.4) with spins of norm $\sqrt{\beta}$. Its per-bond coupling is β. The accompanying identity $S_1 \cdot S_2 = \operatorname{tr}(\varphi(S_1)^{-1}\varphi(S_2))$ [7, p. 131] holds for the normalised trace $\frac{1}{2}\operatorname{Tr}$. Read with $\operatorname{tr} = \chi$, it is off by a factor 2, which accounts for the discrepancy.

Accordingly, the stated range $\beta < 1/(2(\nu - 1))$ [7, pp. 133, 147], obtained in their Sect. 5 and its Appendix, corresponds to $\beta_W < 1/(2(\nu - 1))$ (1/6 for $\nu = 4$) rather than $\beta_W < 1/(\nu - 1)$. The "more refined estimate" extending to $\beta < 1/(\nu - 1)$, stated without proof [7, p. 147], corresponds at face value to $\beta_W < 2/3$ for $\nu = 4$. That is the range of Theorem A.

The reduction used in the present paper (Lemma 2.2) has per-bond coupling $K = \beta_W$, consistent with [2].

8 Discussion

(1) *Beyond* 2/3. By Theorem C(iii), passing 2/3 on the slab requires block (Dobrushin–Shlosman) or multiscale criteria; whether these can do so is open. By Theorem C(iv), no slab-based argument passes $K_c \approx 0.936$. (2) *Other groups.* The vMF structure on S^3 is special to SU(2). For SU(N), $N \geq 3$, the single-link law is a matrix Langevin law, the support is not the boundary of a ball, and the Neumann-flow argument does not transfer. (3) *Scope.* All statements are at fixed lattice spacing, deep in strong coupling; the SU(2) crossover is near $\beta_W \approx 2.2$. Nothing here concerns the continuum limit.

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A Reproducibility

Scripts (Python with mpmath and sympy). The certificates of Lemma 3.4 use exact rational arithmetic throughout, including the tail majorants; mpmath is used only for consistency checks:

- `cert_series.py`: exact series coefficients of F_B and D_A and the partial sums in Lemma 3.4;
- `cert_tails.py`: majorant tail bounds;
- `track1_closedforms.py`: Proposition 3.3 against direct quadrature;
- `thmC_BE_check.py`, `thmCD_checks.py`: Theorem C(i)–(ii) and the MAD grid for Conjecture 6.2;
- `df80_check.py`: the identity in Remark 7.1.

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